# SOP-IT-004 · Tripwires, firewalls and shields: every layer between a keystroke and a public page

SOP IT-004 | Status draft | Changed 2026-11-04 | For Anyone who commits, merges or clones

**Department:** IT · **ID:** 004 · **Status:** draft (approved when tagged) · **Owner:** PIR®'s stewards · **Last changed:** 2026-11-04

> ***Note.*** *Written for a newcomer who has heard the words tripwire, allowlist, attestation, deploy key and hook and wants one page that says what each one catches, where it sits, and what it cannot do. Nothing here needs a terminal to understand. The pages it points to say how to run each piece.*

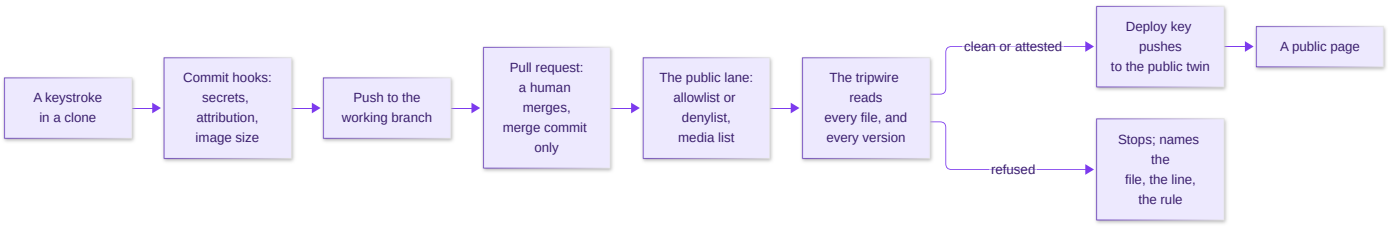
## Purpose

Name every guard in the drasticstatic ecosystem, in the order a change meets them, so that a person who clones one of our repositories knows why it is safe to fetch, a person who merges knows what has already been checked, and a person who finds a refusal knows which guard spoke and what it wants.

## Scope

Every repository the lab tends: the labs under drasticstatic, the organisations' desks, rooms and twins, and partners' organisations built on the same template. It describes the guards as they stand on the date above; each guard's own page is the source for its details.

## The path, and the guard at each step




## The guards, one by one

| # | Guard | Where it sits | What it catches | What it cannot do | Its page |
|---|---|---|---|---|---|
| 1 | **ggshield** (secret scan) | the `pre-commit` hook on your own machine | API keys, tokens, private keys about to be committed | run on a machine that never installed the hooks; see #10 | `my-template/walkthroughs/SECRET-SCAN.md` |
| 2 | **Commit attribution hook** | the `commit-msg` hook | a commit without the two trailers (who wrote it, which session) | prove the trailer is true; it checks shape | `my-template/AGENT-SYNC/README.md` |

| | | | | | |
|---|---|---|---|---|---|
| 3 | **Image guard** | the same `pre-commit` hook, and `image-size-check.yml` on pull requests | a picture over 15 MB (refused), over 2 MB (warned, with the shrink command) | judge whether a picture should be public at all; that is #6 | `my-template/.claude/skills/compress-images.md` |
| 4 | **The merge rule** | GitHub, per repository: squash and rebase turned off | history rewritten by a squash; a merge by a helper (helpers stop at the merge) | force a review on a private repository without a paid plan (see *Safe to merge*) | SOP-OPS-002 |
| 5 | **Allowlist** (rooms, twins) or **denylist** (labs) | `PUBLIC-ALLOWLIST.txt`, or the lane's strip list in `sync-public.yml` | a path that was never listed (allowlist) or was listed as private (denylist) leaving at all | read inside a file; that is #7 | SOP-OPS-001, SOP-OPS-003 |
| 6 | **Media list** | `PUBLIC-LANE-MEDIA.txt` | a picture or font outside a folder a person approved by name and date | tell a good picture from a bad one; it trusts the person's yes | SOP-OPS-003 |
| 7 | **The tripwire** (`scan-public-lane.sh`) | the lane, just before the push, on the current tree and on every historical version | credentials (never pass), identifiers (SSN, EIN, account numbers), street-address shapes, giver-list and personal-matter words, legal-lane folders, unreadable binaries, oversize files | know intent: a public shop address and a home address look alike, which is why #8 exists | `my-template/scripts/scan-public-lane.sh`, `TRIPWIRE-COPIES.md` |
| 8 | **Attestation** | `PUBLIC-ALLOWLIST-ATTESTED.txt`, one line per file, signed by an officer, fingerprinted | nothing; it opens a refusal a person has read and accepts (a shop address, a company's line in a statement) | open a credential; those stay refused whatever the line says | SOP-OPS-003 |
| 9 | **Deploy keys** | one key pair per lane: private half as a secret on the source, public half on one twin | a lane pushing anywhere but its own twin; a stolen token with wide reach | protect the twin from a person with write access to it | `DEPLOY-KEYS_20261102.md` on the ARB desk; SOP-OPS-003 rotation |
| 10 | **Safe to clone** | repository policy, `SECURITY.md`, and | a `package.json` that runs a script on install; hooks that run | protect a machine that runs `npm` | this page, *Safe to clone* |

| | | | | | |
|---|---|---|---|---|---|
| | | `lifecycle-scripts-check.yml` | without being installed by hand | `install` on a stranger's fork; that is #13 | |
| 11 | **Dependabot** and **Dependency Review** | GitHub | a dependency with a known vulnerability; a pull request that adds one | fix it; a person merges the bump | `my-template/walkthroughs/DEPENDABOT.md` |
| 12 | **The safety bundles** and **Time Machine** | `git-safety-backup.sh` to iCloud; the Time Machine drive | work lost to a dead machine: uncommitted edits, stashes, unpushed branches | replace a push; GitHub is the record | `my-template/walkthroughs/ICLOUD-AND-GITHUB-TOGETHER.md`, proposal 019 |
| 13 | **The evaluation VM** | a virtual machine on the Studio or the iMac | a fork or a tool under review doing harm to the real machine | review the code for you; the checklist does | proposal 013; the external-repo checklist in every `CLAUDE.md` |
| 14 | **Session messaging limits** | `isolatePeerMachines`, per-repo `crossSessionInbound` | a session's doorbell leaving the machine or reaching a walled-off lane | replace the written guideline that says who may ring whom | `my-template/walkthroughs/SESSION-MESSAGING.md` |

## Safe to clone: why fetching one of our repositories is safe

1. **Nothing runs when you clone or install.** Our `package.json` files carry no `preinstall`, `postinstall` or `prepare` scripts, and `lifecycle-scripts-check.yml` fails any pull request that adds one. Git hooks live in `.githooks/` and do nothing until a person runs `sh scripts/install-hooks.sh`.
2. **What the hooks do is written down** (#1, #2, #3 above) and each is a short shell script you can read before installing.
3. **Every public page was read before it left** (#7), and the private history never travels: the twin is rebuilt from the listed tree.
4. **A `SECURITY.md`** in each repository says the above in one paragraph and names where to report a concern.

## Safe to merge: what a stranger's pull request meets

1. The checks run on the pull request itself: the tripwire self-test, ggshield, the image guard, the lifecycle-script check, Dependency Review. A red check blocks the merge button for anyone who respects it.
2. A person with the role merges, with a merge commit, never a helper (SOP-OPS-002).
3. **An honest limit:** GitHub enforces "required checks" and "required reviews" through branch protection or rulesets, which are free on public repositories and need GitHub Pro (personal) or Team (organisations) on private ones. On a private repository without that plan, step 2 is a rule people keep, not a lock the platform holds. The stewards' page for each repository says which it is.
4. Code from outside is read in the evaluation VM first (#13) when it is more than a page.

## When a guard speaks

| You see | Who spoke | Do |
|---|---|---|
| "No secrets have been found" | ggshield | nothing; carry on |

| | | |
|---|---|---|
| "ggshield found …" | ggshield | stop; the secret never enters a commit; rotate it if it was ever real |
| "refused … a street-address shape (line N)" | the tripwire | open the line; a public business address gets an attestation from an officer (#8); a person's home address gets reworded |
| "refused … over 15 MB" | the tripwire | shrink the file (#3); the lane cannot be talked into it |
| "a legal lane … appears in the history" | the tripwire | the folder's old name is still in history; the lane renames it in the published history (`--path-rename`) or holds the path |
| "secret … is not set, skipping push" | the lane | the deploy key was never minted; run the lane's `setup-lane-keys.sh` as the owner |
| a grey merge button | GitHub | you are not an owner on this repository, or a check is red; read the checks |

## Roles

| Role | Does |
|---|---|
| Contributor | installs the hooks once, commits with the trailers, opens pull requests, never merges their own |
| Officer or steward | merges, signs attestations, approves media folders by name and date, mints deploy keys |
| Helper (an agent) | everything up to the merge; never an attestation, never a merge, never a public-surface action without a person's command |

*Sources: the pages named in the table, all in PIR®'s procedures or in `my-template`, read on 2026-11-04. Not legal advice.*